

Chapter 1 Warm-Up: Python Lists and Tuples

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1 Python Lists

1. A Python **list** is an ordered collection of objects. Lists are written using square brackets, `[ ]`, with the elements separated by commas.

- Begin by creating a list containing three student names:

```
students = ["Alice","Bob","Carlos"]
```

- Print the entire list:

```
print(students)
```

- Individual elements of a list can be accessed using their position, or *index*. Python begins counting at zero. Try:

```
print(students[0])
print(students[1])
print(students[2])
```

- Which student is stored at index 0? Which student is stored at index 2?

2 Lists Inside Lists

1. The elements of a list do not have to be simple strings or numbers. A list can itself contain another list.

Suppose we want to represent a course together with its students.

- Create a list containing the name of a course and the list of students:

```
course = ["Python Programming",students]
```

- Print the complete object:

```
print(course)
```

- The first element of `course` is the course name:

```
print(course[0])
```

- The second element is itself a list containing the students:

```
print(course[1])
```

- Because `course[1]` is a list, we can select an element from that list as well. Try:

```
print(course[1][0])
print(course[1][2])
```

3 Python Tuples

1. A Python **tuple** is another ordered collection of objects. Tuples are similar to lists, but are written using parentheses, `( )`.

- Create two simple course lists:

```
course1 = ["Python Programming",["Alice", "Bob"]]
course2 = ["Calculus I",["Diana", "Ethan"]]
```

- Combine the two course lists into a tuple:

```
courses = (course1,course2)
```

- Print the complete tuple:

```
print(courses)
```

- Access the first and second course using:

```
print(courses[0])
print(courses[1])
```

- Print only the name of the second course:

```
print(courses[1][0])
```

- Print the name of the first student enrolled in the second course:

```
print(courses[1][1][0])
```

4 Warm-Up Exercise

Construct a Python object representing two laboratory sections. Each laboratory section should contain a course name and a list of three students. Use the following information:

Physics Laboratory

Maya
Noah
Olivia

Computer Laboratory

Liam
Sophia
Ethan